

PERFORMANCE INTELLIGENCE REPORT

A Century of Football

Decoding National Team Performance, 1872–2026

Task 2.1 • AI Team Training '27 • Phase I

Data: International Football Results (1872–2026) — results.csv, goalscorers.csv, shootouts.csv

Executive Summary

This report analyzes 49,518 international matches (1872–2026), merging results, goal events, and shootout records into one view. Five questions drove the analysis: who dominates on raw output, who is actually efficient once small-sample noise is removed, when shootout drama peaks, which teams squander leads, and how the game has changed era by era.

England, Germany and Brazil lead all-time goals; Cristiano Ronaldo leads individual scorers. Raw efficiency rankings are nearly meaningless without a minimum-matches filter — and even filtered, two small federations (Jersey, Guernsey) still outrank major nations, since match volume alone doesn't correct for opponent strength. Shootouts have grown steadily since the 1980s and are far more common in knockout-heavy regional competitions than in flagship tournaments. Goals per match roughly halved between the 1880s and 1980s before stabilizing into a more defensive modern game.

Methodology

- **Cleaning:** dropped 2 unplayed/scoreless fixtures; trimmed whitespace in team/tournament names; disambiguated 2 same-day duplicate matches with a sequence key.
- **Merging:** `goalscorers.csv` and `shootouts.csv` carry no tournament/decade info — both merged onto `results.csv` on (date, home_team, away_team); 682/683 shootouts matched.
- **Feature engineering:** decade, goal margin, match outcome derived from raw scores; results reshaped into one row per team per match (“long format”) for simple team-level aggregation; standard points (3/1/0) computed per team per match.

1 Top Performers

Which teams have scored the most goals in history? Who are the top scorers?

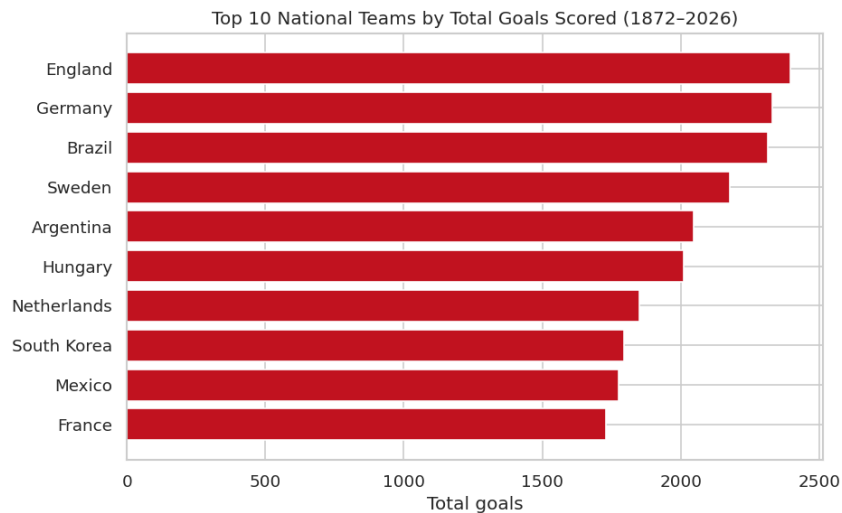


Figure 1: Top 10 national teams by total career goals.

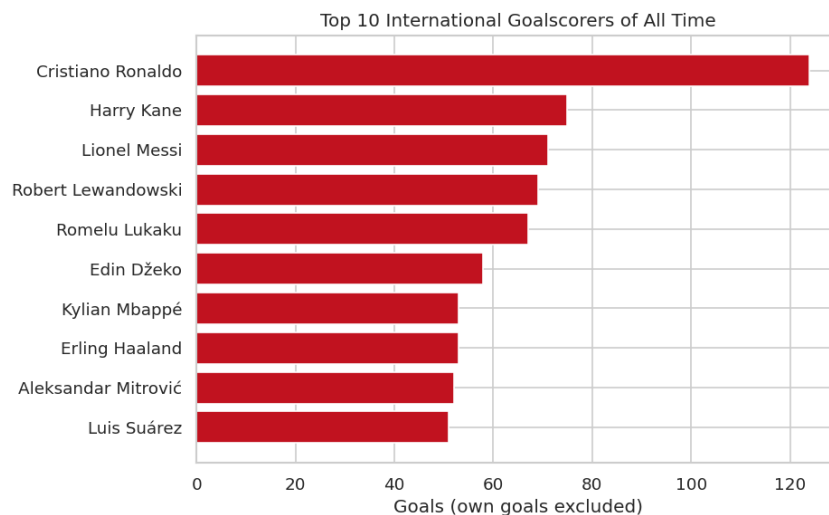


Figure 2: Top 10 individual goalscorers (own goals excluded).

England leads with 2,395 career goals, narrowly ahead of Germany (2,331) and Brazil (2,315) — a lifetime total that favors nations with the longest fixture histories. Cristiano Ronaldo’s 124 goals lead individually, well clear of Harry Kane (75) and Messi (71). Own goals were excluded — credited to the benefiting team in this dataset, not a goal the scorer would claim.

2 Team Efficiency

Which team earns the most points/goals per match? Does everyone in the raw top 10 belong there?

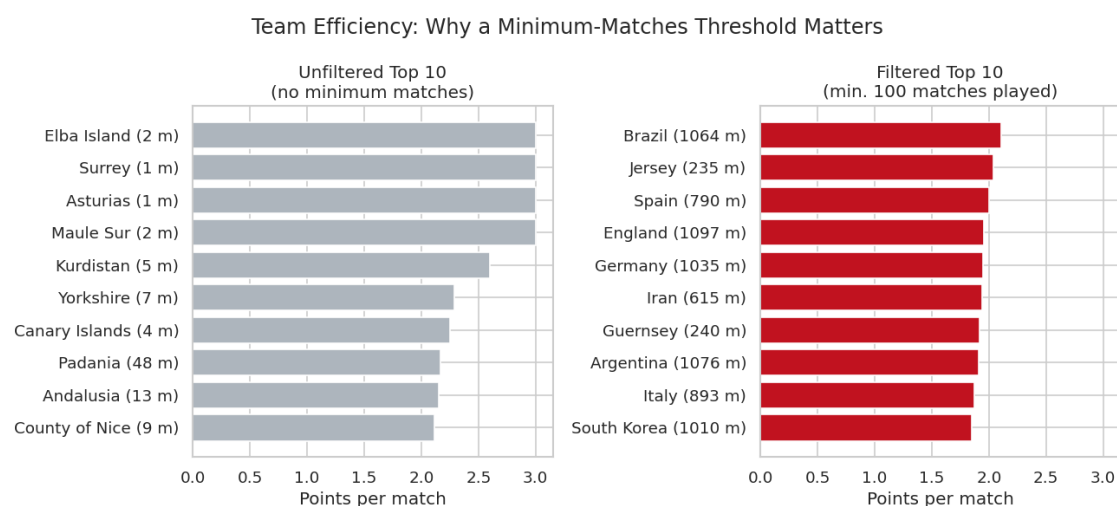


Figure 3: Points-per-match leaders, with and without a minimum-matches filter.

Unfiltered, the top spots go to teams with 1–5 lifetime matches (Surrey, Elba Island, Asturias — all a perfect 3.00 points/match on almost no games). A 100-match minimum removes 127 of 337 teams and produces:

Table 1: Team Efficiency — Filtered Top 10 (min. 100 matches)

Team	Matches	Pts/Match	Goals/Match
Brazil	1,064	2.11	2.18
Jersey	235	2.04	2.74
Spain	790	2.01	2.04
England	1,097	1.96	2.18
Germany	1,035	1.95	2.25
Iran	615	1.94	1.88
Guernsey	240	1.92	2.57
Argentina	1,076	1.91	1.90
Italy	893	1.87	1.75
South Korea	1,010	1.85	1.78

Brazil tops it, as expected — but Jersey (#2) and Guernsey (#7) are not FIFA members; they play mostly other small regional sides, not the world’s strongest teams. A matches threshold removes small-sample noise but does not correct for strength of schedule.

3 Drama Analysis — Penalty Shootouts

Which decade produced the most shootouts? Which tournaments cause them most often?

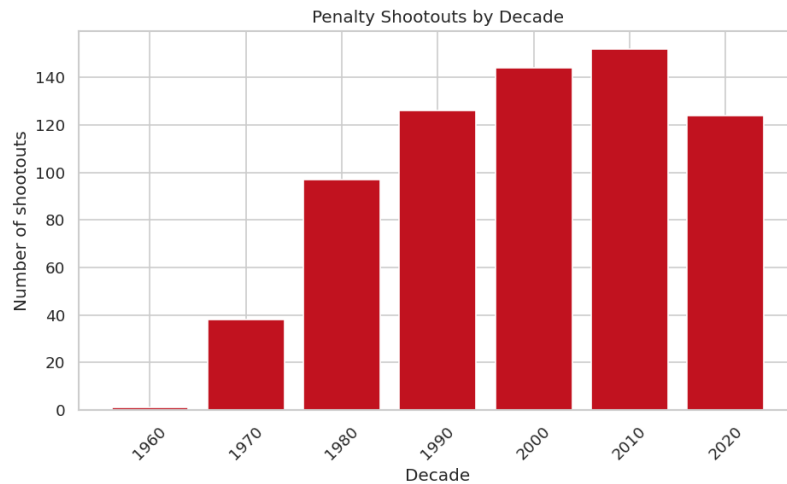


Figure 4: Penalty shootouts by decade — peaking at 152 in the 2010s.

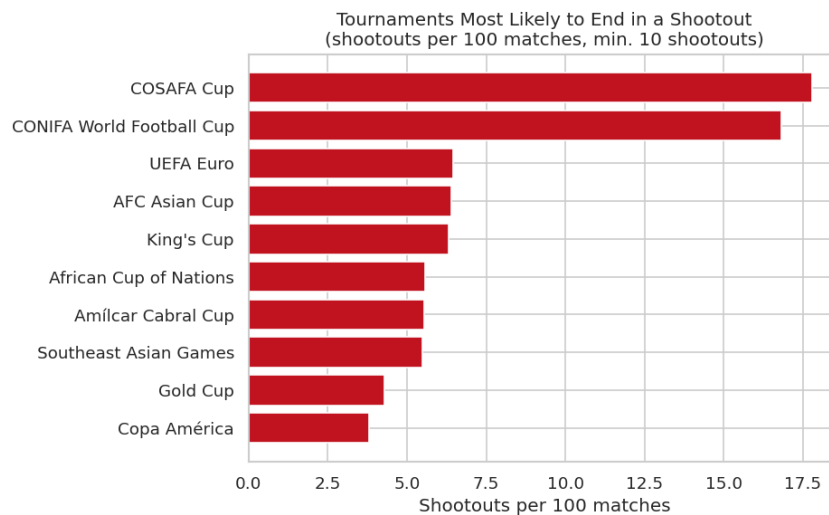


Figure 5: Shootout rate — shootouts per 100 matches (min. 10 shootouts).

By raw count, Friendlies “cause” the most shootouts (79) — but they’re 37% of all matches, so that’s just volume. By **rate**, the COSAFA Cup (17.8%) and CONIFA World Football Cup (16.8%) are far more shootout-prone than flagship events like the UEFA Euro (6.4%) or Africa Cup of Nations (5.6%) — both use heavily knockout-structured, short-tournament formats.

4 Worst Performers — Squandered Leads

Which teams most often score first and still fail to win?

Determining “scored first” needs reliable goal-minute data; only 15,466 of 49,518 matches (31.2%) were usable (0–0 draws have no first goal by definition, and any match with a missing goal-minute was excluded to avoid guessing at order).

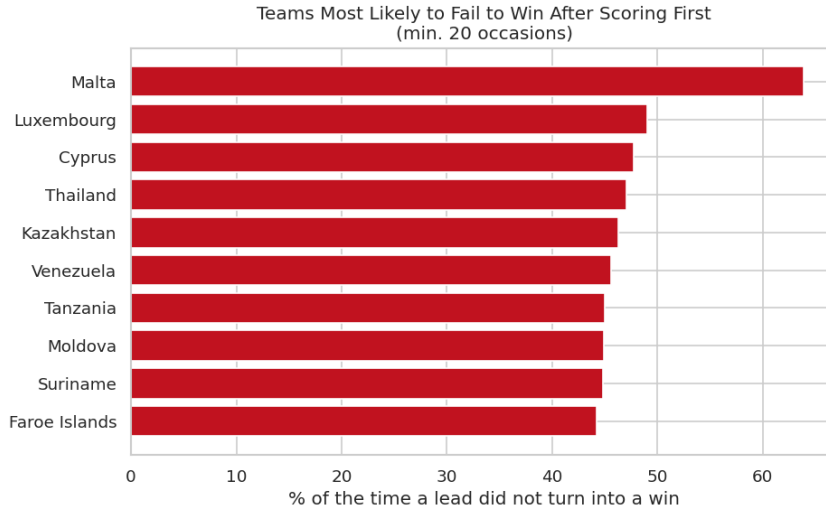


Figure 6: Highest rate of failing to win after scoring first (min. 20 occasions).

Table 2: Worst Performers by Rate (min. 20 occasions scoring first)

Team	Scored First	Failed to Win	Rate
Malta	36	23	63.9%
Luxembourg	51	25	49.0%
Cyprus	65	31	47.7%
Thailand	51	24	47.1%
Kazakhstan	54	25	46.3%

Malta fails to convert a lead into a win nearly two-thirds of the time — the opposite profile from Brazil (17.4% fail rate on 328 occasions). Weaker sides taking a shock lead against stronger opposition tend to get overrun; elite teams that score first are usually already the better side.

5 Era Comparison

How have goals/match, draw rates, and winning margins changed decade by decade?

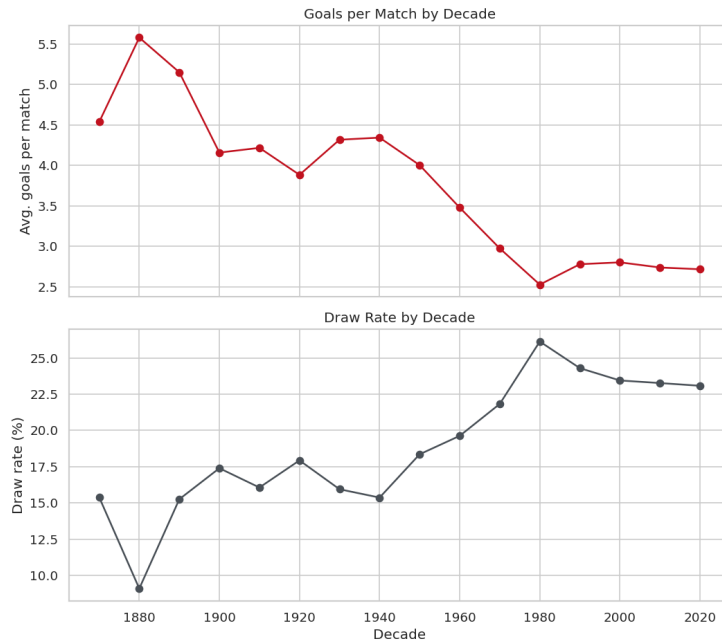


Figure 7: Goals per match and draw rate by decade.

Goals per match fell from a 19th-century peak of 5.58 (1880s) to a low of 2.53 (1980s) — a 54% decline — before stabilizing around 2.7–2.8 since the 1990s. Draw rate roughly doubled over the same span (9–17% early on vs. 23–26% since the 1980s); winning margins shrank from 3–4+ goals to about 2. Early decades have small samples (13 matches in the 1870s) and should be read directionally.

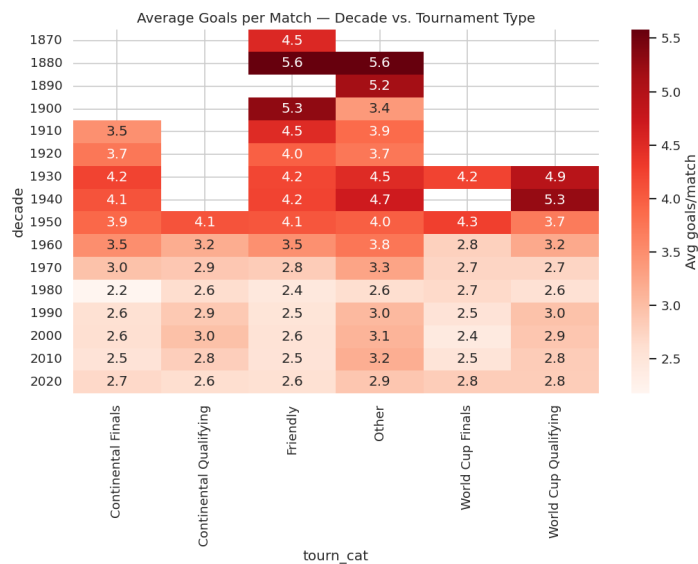


Figure 8: Average goals per match by decade and tournament type.

The decline holds across every tournament type — Friendlies, qualifiers, and Continental finals all follow the same downward path, confirming this is a change in how the game is played, not a dataset artifact.

6 Outlier Detection — Most Lopsided Matches

Table 3: Top 5 Most Lopsided Matches on Record

Date	Home	Away	Score	Tournament
2001-04-11	Australia	American Samoa	31–0	FIFA WC qualification
1971-09-13	Tahiti	Cook Islands	30–0	South Pacific Games
1979-08-30	Fiji	Kiribati	24–0	South Pacific Games
2013-06-24	Provence	Tibet	22–0	Intl. Tournament of Peoples
2001-04-09	Australia	Tonga	22–0	FIFA WC qualification

Australia’s 31–0 win over American Samoa remains an official FIFA world-record scoreline. A z -score check (margin > 6 std. above the historical mean of 1.72) flags 95 matches (0.19%) as statistically extreme; several involve non-FIFA regional/cultural teams (Tibet, Sápmi, Padania) rather than official national sides — genuine results, not data errors, but worth filtering for official benchmarking.

7 Insights & Recommendations

- **Coaching:** the modern game is genuinely more defensive and tighter than any era before 1990 — game plans should assume low-scoring, close matches as the norm.
- **Coaching:** teams with a high squandered-lead rate (Malta, Luxembourg, Cyprus) may benefit from targeted lead-protection coaching, since this is a recurring pattern, not noise.
- **Federations:** any ranking built from this data needs both a minimum-matches filter *and* a strength-of-schedule adjustment — Jersey/Guernsey show a matches filter alone isn’t enough.
- **Federations:** prepare specifically for penalty shootouts ahead of knockout-heavy regional tournaments (COSAFa-style formats), given the roughly 1-in-6 rate.
- **Media:** the 1880s–1980s scoring decline is a well-supported long-run story, though pre-1900 decades rest on small samples.

8 Limitations & Data Notes

- `goalscorers.csv` coverage is partial — the “scored first” analysis used only 31.2% of matches.
- Dataset includes non-FIFA regional/cultural teams (Jersey, Guernsey, Tibet, Padania, Sápmi, Darfur) alongside official national teams; not always comparable.
- 2 fixtures with no final score were excluded from all analysis.
- All thresholds (100 matches, 20 occasions, 6 std. dev.) are reasonable defaults, not universal standards — tunable by audience.